Circuit électrique

Série N°1 :

EX1 :

1) $m_m = m_B^{6} - m_m^{4} + 1 = 3 \Rightarrow$ 3 mailles indépendantes.

2) maille 1 : $E - R_3 I_3 - R_2 I_2 - R_6 I_6 = 0$
maille 2 : $R_2 I_2 - R_3 I_3 - R_4 I_4 = 0$
maille 3 : $R_6 I_6 + R_4 I_4 - R_5 I_5 = 0$

3) A. $I_1 = I_2 + I_3$ B. $I_1 = I_5 + I_6$

C. $I_6 = I_2 + I_4$ D. $I_3 = I_4 + I_5$

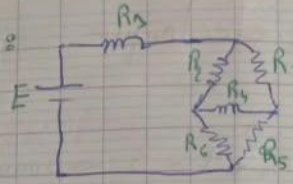
4)
$$\begin{cases} -R_2 I_4 + R_3 I_5 + (R_3 + R_2 + R_6) I_6 = E \\ -(R_2 + R_3 + R_4) I_4 - R_3 I_5 + R_2 I_6 = 0 \\ R_4 I_4 - R_5 I_5 + R_6 I_6 = 0 \end{cases}$$

5)
$$\begin{pmatrix} -R_2 & R_4 & R_4+R_2+R_6 \\ -(R_2+R_3+R_4) & -R_3 & R_2 \\ R_4 & R_5 & R_6 \end{pmatrix} \cdot \begin{pmatrix} I_4 \\ I_5 \\ I_6 \end{pmatrix} = \begin{pmatrix} E \\ 0 \\ 0 \end{pmatrix}$$

Δ

$$I_4 = \frac{\Delta I_4}{\Delta} \qquad \Delta I_4 = \begin{vmatrix} E & R_4 & R_4+R_2+R_6 \\ 0 & -R_3 & R_2 \\ 0 & R_5 & R_6 \end{vmatrix}$$

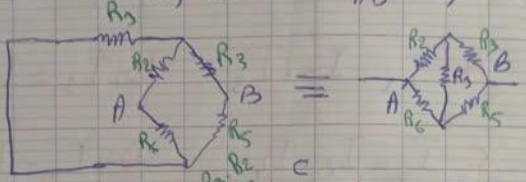
Ex2 :

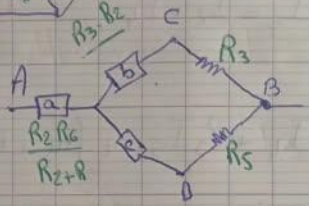


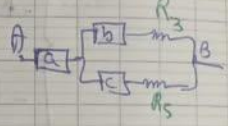
Thevenin

1) Débranche R_4 2) Court circuit

3) calculer R_{AB} 4) rebranche E et calcule $E_{th} = U_{AB}$







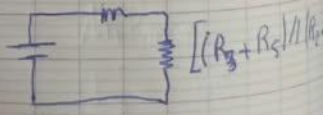
$$R_{AB} = a + [(b+R_3) /\!/ (c+R_6)]$$



$$E_{th} = U_3 - U_2$$
$$= R_3 I_{03} - R_2 I_{02}$$

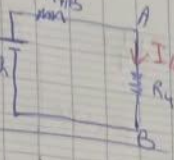
$$I_2 = \frac{(R_3+R_5)\,I_{03}}{R_2+R_3+R_5+R_6} \quad ; \quad I_{03} = \frac{R_2+R_6}{R_2+R_3+R_5+R_6}$$

$$I_{03} = \frac{E}{R_3 + [(R_3+R_5) /\!/ (R_2+R_6)]}$$



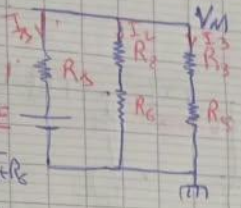
$$\begin{cases} E - U_1 - U_2 - U_6 = 0 \\ U_6 + U_2 - U_3 - U_5 = 0 \\ 2I_0 + (R_2+R_6)\,I_{02} = E \\ -R_3+R_5 = I_{03}\,(R_2+R_6)\,I_{02} = 0 \end{cases}$$

$E_{TH} = R_3 I_{03} - R_2 I_{02}$



$I_4 = \frac{E_{TH}}{R_{TH} + R_4}$

Millman



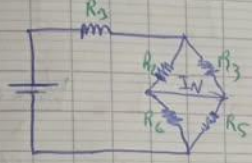
$V_M = \frac{\frac{E_1}{R_3} + \frac{0}{R_2+R_6} + \frac{0}{R_1+R_5}}{\frac{1}{R_3} + \frac{1}{R_2+R_6} + \frac{1}{R_1+R_5}}$

$I_{02} = \frac{V_M - E}{R_2 + R_6}$

$I_{03} = \frac{V_M - E}{R_3 + R_5}$

$I_{01} = \frac{V_M - E}{R_1}$

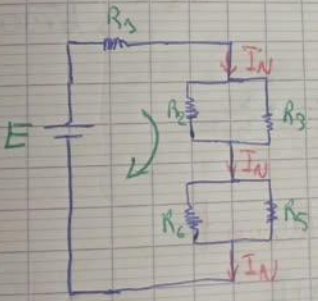
Norton



$U_{AB} = V_A - V_B = 0$

↺ Millman

$I_4 = \frac{U_{TH}}{R_{TH} + R_4}$



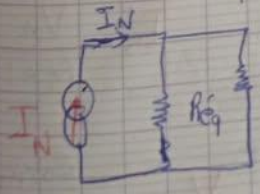
$I_4 = \frac{R_{AB} I_N}{R_{AB} + R_4}$

Rebranche G
Court circuit AB
et Calculer $I_{AB} = I_N$

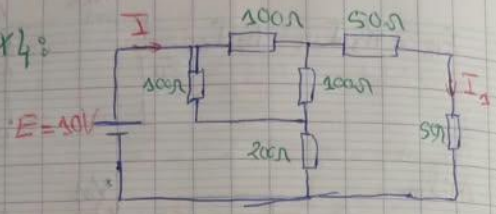
$I_N = \frac{E}{R_1 + (R_2 // R_3) + (R_5 // R_6)}$

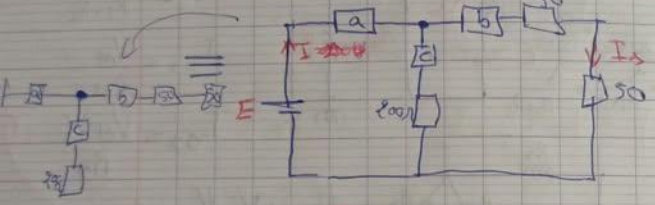
$I_4 = \frac{R_{éq} \cdot I_N}{R_{éq} + R_4}$

$R_{éq} = R_{TH}$



EX4 :





Par ~~M~~ Milman : $I = \dfrac{V_M - E}{a}$

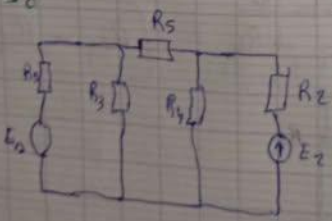
$$I_5 = \frac{V_M}{b+10}$$

$$V_M = \frac{E/a}{1/a + \frac{1}{c+200} + \frac{1}{b+100}}$$

Par diviseur de courant :

$$I_5 = \frac{(c+200)I}{c+200+b+100} \qquad I = \frac{E}{a + [(c+200)//$$

EX3 :



Milman :

$$I'_5 = \frac{V_M}{R_5 + (R_4 // R_6)}$$

$$V_M = \frac{E_1/R_1}{\frac{1}{R_1} + \frac{1}{R_3} + \frac{1}{R_4} + \frac{1}{R_6}}$$

Diviseur de Courant : $I'_5 = \frac{R_3 I'_3}{R_3 + R_5 + (R_4 // R_2)}$

$$I'_3 = \frac{E_3}{R_3 + R_3 // (R_5 + (R_2 // R_4))}$$